Machine Learning Sequential Data Report

AvaRaisian

1 Report Task 5: Posterior Summaries and $\hat{R}$ Diagnostics

1.1 Added Code

The following code was added after MCMC sampling to generate posterior summaries and $\hat{R}$ diagnostics:

```
summary = az.summary(idata, round_to=3)
print(summary)
```

1.2 Reported $\hat{R}$ Values

All parameters exhibit reliable convergence diagnostics:

- Intercept: $\hat{R} \approx 1.006$
- Month-level effects month_mu[0–3]: $\hat{R} \approx 1.001$–1.003
- Latent means $\mu[t]$: $\hat{R} \approx 0.999$–1.002

1.3 Why $\hat{R}$ Is Important

$\hat{R}$ compares between-chain and within-chain variance to assess mixing. Values close to 1 indicate stable sampling and trustworthy posterior estimates, while values significantly above 1.1 signal poor convergence and unreliable inferences.

1.4 Model Misfit Explanation

The model assumes deaths follow a smooth trend with stable seasonality. That structure works well for the pre-COVID era, and if COVID had not occurred, the forecasts would likely remain accurate. The large post-2020 errors reflect structural misfit, not MCMC failure: the model simply cannot represent the abrupt mortality shock introduced by the pandemic.

2 Report Task 6: Effect of Using Pre-2010 Data

The only modification to the code:

```
pre = df[df.index < "2010"]
post = df[df.index >= "2010"]
```

2.1 Posterior Summaries for Pre-2010

Below are the trace plots and posterior distributions obtained after re-running MCMC on the pre-2010 data.

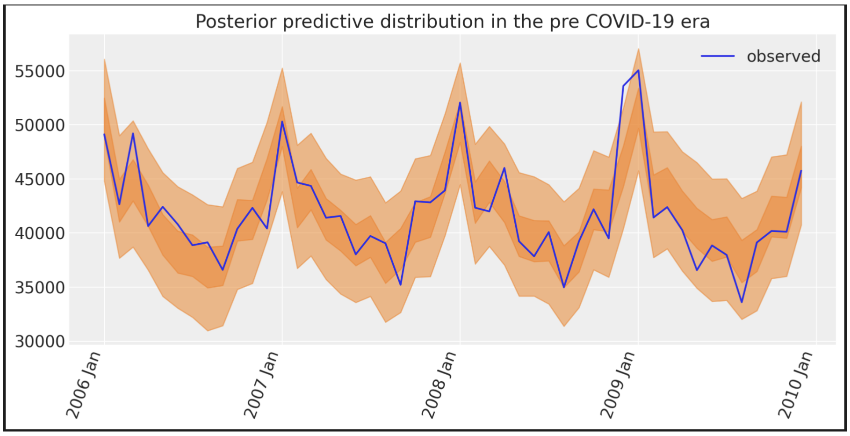
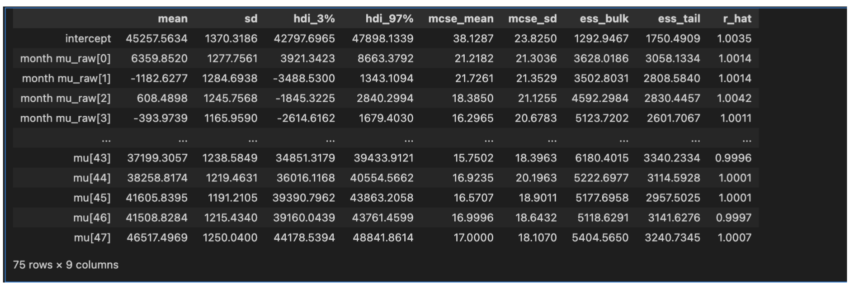


Figure 1: Posterior predictive distribution (pre-2010).



	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
intercept	45257.5634	1370.3186	42797.6965	47898.1339	38.1287	23.8250	1292.9467	1750.4909	1.0035
month mu_raw[0]	6359.8520	1277.7561	3921.3423	8663.3792	21.2182	21.3036	3628.0186	3058.1334	1.0014
month mu_raw[1]	-1182.6277	1284.6938	-3488.5300	1343.1094	21.7261	21.3529	3502.8031	2808.5840	1.0014
month mu_raw[2]	608.4898	1245.7568	-1845.3225	2840.2994	18.3850	21.1255	4592.2984	2830.4457	1.0042
month mu_raw[3]	-393.9739	1165.9590	-2614.6162	1679.4030	16.2965	20.6783	5123.7202	2601.7067	1.0011
...	...	...	...	...	...	...	...	...	...
mu[43]	37199.3057	1238.5849	34851.3179	39433.9121	15.7502	18.3963	6180.4015	3340.2334	0.9996
mu[44]	38258.8174	1219.4631	36016.1168	40554.5662	16.9235	20.1963	5222.6977	3114.5928	1.0001
mu[45]	41605.8395	1191.2105	39390.7962	43863.2058	16.5707	18.9011	5177.6958	2957.5025	1.0001
mu[46]	41508.8284	1215.4340	39160.0439	43761.4599	16.9996	18.6432	5118.6291	3141.6276	0.9997
mu[47]	46517.4969	1250.0400	44178.5394	48841.8614	17.0000	18.1070	5404.5650	3240.7345	1.0007

75 rows × 9 columns

Figure 2: Monthly seasonal effect (μ_m) posterior for pre-2010.

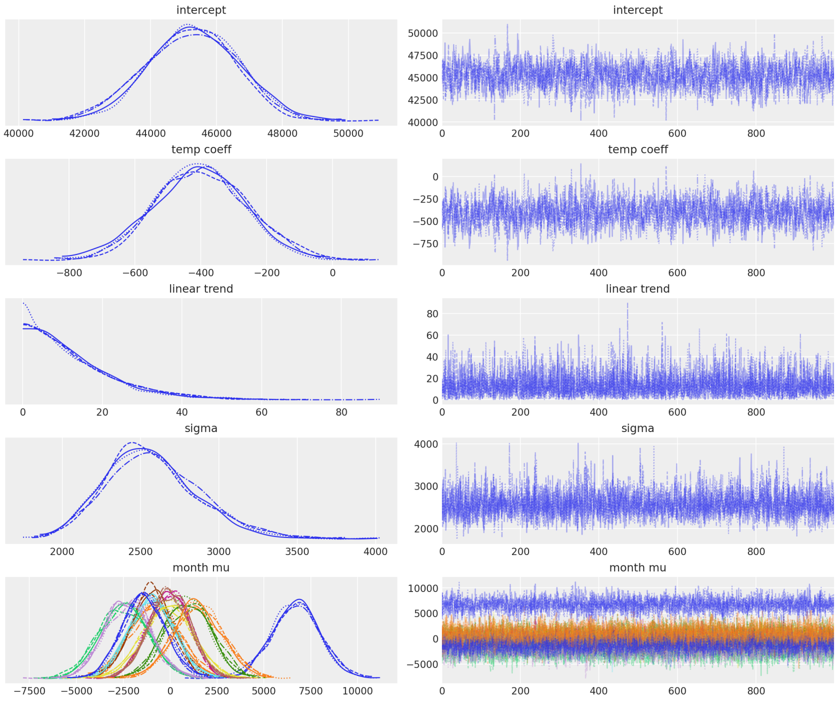


Figure 3: Posterior densities and trace plots for pre-2010 parameters.

2.2 Comparison With Original Pre-2020 Posteriors

Training on pre-2010 data lowers the intercept, weakens seasonal effects and largely removes the linear trend because early years have lower deaths, less variability, and too short a time span to identify long-term structure. Parameter uncertainty also increases. In contrast, the pre-2020 model shows stronger trends, larger seasonal amplitudes, and much tighter posteriors.

3 Report Task 7: Posterior Predictive Checks

The full model predicts training data well because seasonality and temperature explain most variation. Removing month indicators deletes seasonal patterns, so peaks flatten and winter/summer differences disappear. Removing both month and temperature leaves only a smooth trend, producing the poorest fit: predictions shift toward the overall mean and miss all short-term structure.

Original: With Month Indicators + Temperature

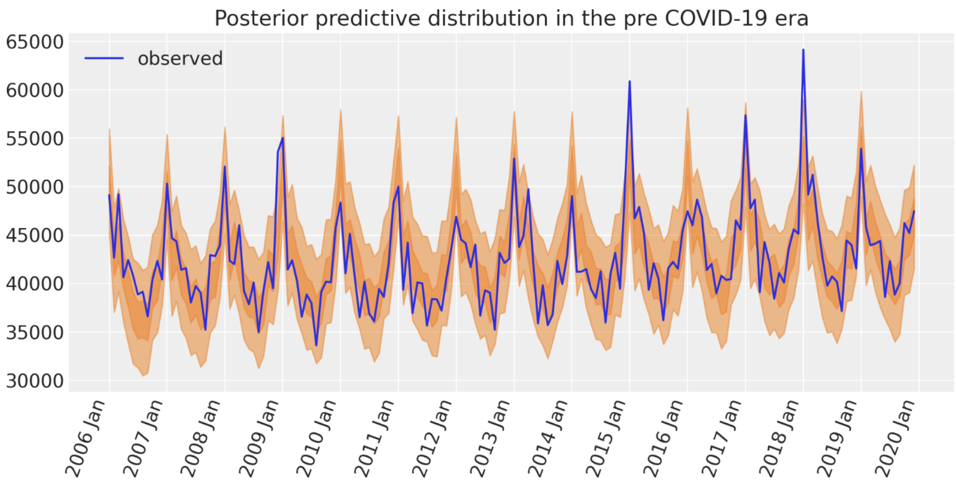


Figure 4: Posterior predictive distribution (original model).

Without Month Indicators

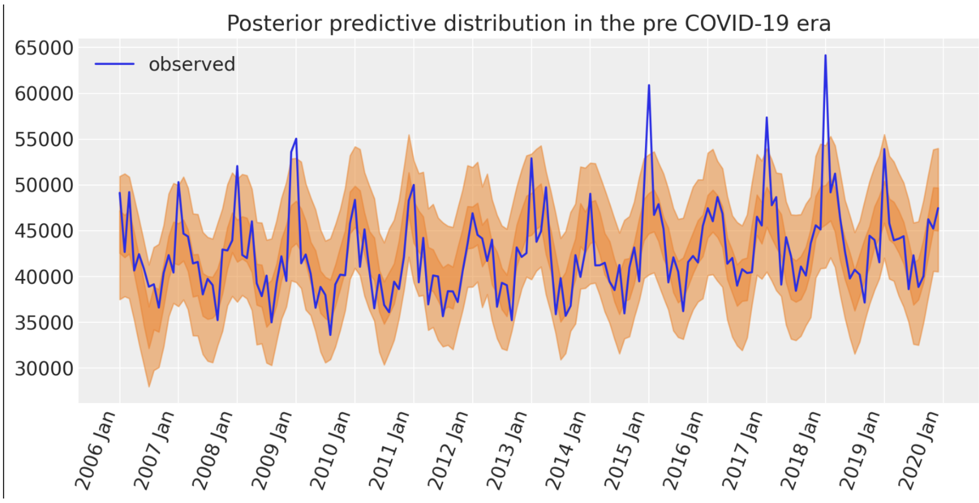


Figure 5: Posterior predictive distribution without month indicators.

Without Month Indicators and Without Temperature

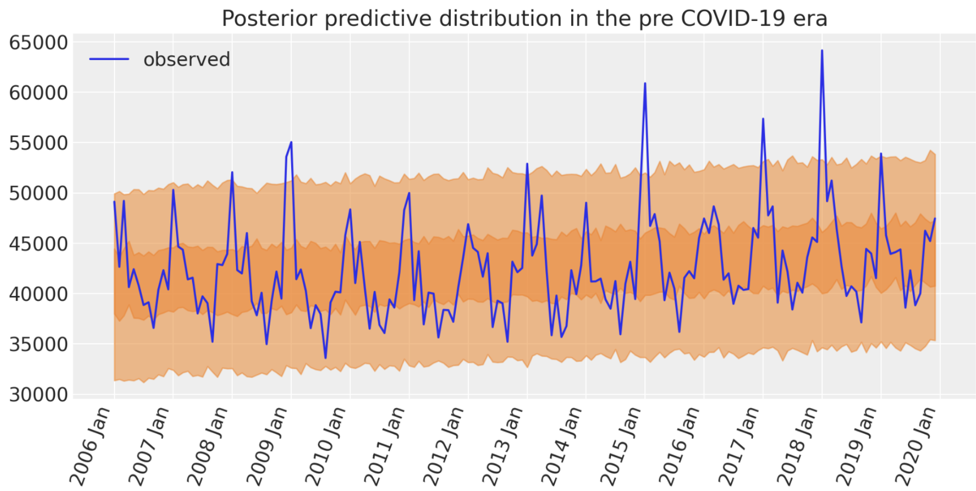


Figure 6: Posterior predictive distribution Without Month Indicators and Without Temperature

4 Report Task 8: HMM Comparison

Using the `sample` method, both HMMs were used to generate synthetic death-category sequences. HMM2 matches the real distribution far better: its KL divergence (596.8 vs. 3076.9), L1 distribution error (0.162 vs. 0.365), and log-likelihood all indicate a closer fit to the underlying data. HMM1 performs poorly because its supervised, frequency-based emission estimates are brittle and overfit one-step counts, producing unnatural samples. Visual inspection confirms that HMM2 reproduces the observed balance of low/medium/high deaths more realistically than HMM1.

Metric	HMM1	HMM2	Better Model
KL Divergence (↓)	3076.94	596.79	HMM2
Distribution Error (L1, ↓)	0.3655	0.1624	HMM2
Log-likelihood (↑)	-195.58	-187.98	HMM2

Table 1: Comparison of sampled HMM sequences against the actual distribution.

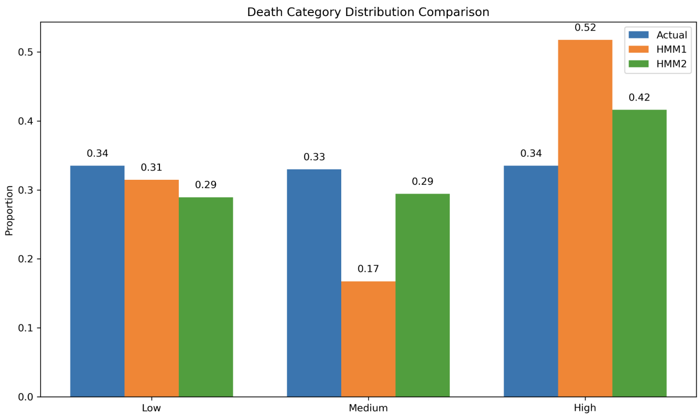


Figure 7: Comparison of Actual and HMM-Sampled Death Category Distributions

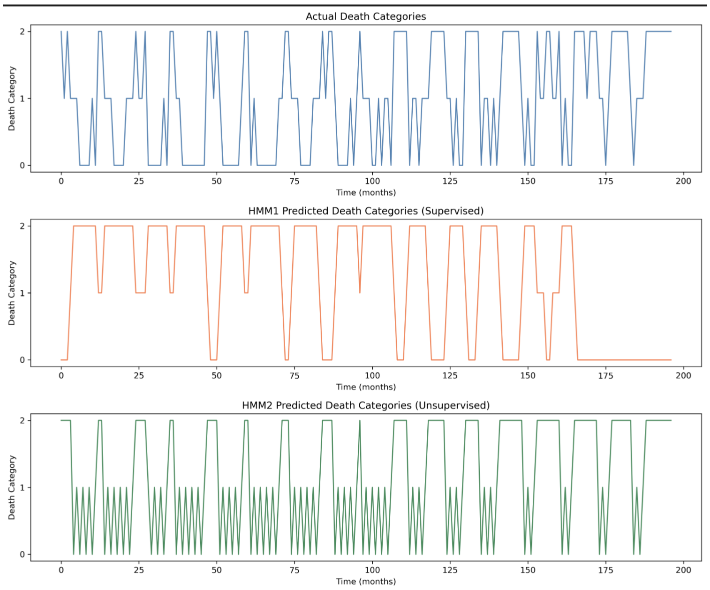


Figure 8: Comparison of Actual and HMM-Sampled Death Category Sequences